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Toxicity and biodegradation of two different hydrothermal liquefaction process waters for anaerobic digestion and the effect of microaeration

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ABSTRACT

Hydrothermal liquefaction (HTL) can convert a considerable portion of the carbon in complex feedstocks into renewable bio-oil, but it also generates an aqueous phase (*i.e.*, HTL process water) that retains >30% of the carbon from the HTL feedstock. Feeding HTL process water to anaerobic digestion (AD) is a promising approach for maximizing resource recovery, enabling the conversion of the retained carbon into biogas. However, various toxic and poorly biodegradable compounds in the HTL process water make its treatment with AD challenging. Presently, the underlying mechanisms remain unclear. We investigated the impact of HTL process water from two different feedstocks – a food-waste proxy (*i.e.*, dog food, rich in proteins) and wheat straw (rich in lignocellulose) – on the different trophic groups in the food web of AD. We found that methanogens, rather than acidogens, were more severely inhibited by HTL process water. Comparative toxicity and biodegradability analyses showed that wheat-straw process water had a higher methane potential of 189 ± 1.52 mL CH_4/g COD than that of food-waste process water with a potential of 130 ± 0.13 mL CH_4/g COD, regardless of its higher toxicity to acetoclastic methanogens. The use of microaeration-acclimated biomass improved the methane yields of food-waste process water by 38% compared to anaerobic biomass, even without oxygen addition. However, the use of microaeration-acclimated biomass or microaeration was ineffective for wheat-straw process water. These findings highlight the importance of feedstock-specific strategies to optimize AD for biogas production from HTL process water.

1. Introduction

Integrating hydrothermal liquefaction (HTL) and anaerobic digestion (AD) is a promising approach for maximizing resource recovery from wet biomass and organic waste [1–3]. HTL operates in hot compressed water (250 °C – 374 °C, 5–20 MPa, 5–60 min) to efficiently convert complex feedstocks into bio-crude oil with higher energy density than the original biomass [4,5]. However, the process generates an aqueous phase retaining >30% of the feedstock carbon, which can be valorized through AD to recover an additional 2%–14% of the initial energy input and reduce its environmental risks [6,7]. Despite this potential, the complex composition of HTL process water (or HTL aqueous phase) poses considerable challenges to AD efficiency.

The inhibitory effects of HTL process water on AD are attributed to byproducts of temperature-driven reactions with macromolecular components in HTL feedstocks. Maillard reactions between amino acids and reducing sugars generate melanoidins and nitrogenous heterocycles (*e.g.*, pyridines, pyrroles) when temperatures exceed 250 °C [8], which are known to disrupt microbial membranes and inhibit methanogenic enzymes [9]. Protein-rich feedstocks yield N-heteroaromatics *via* ammonolysis and cyclization [10], whereas lignocellulose-rich feedstocks yield phenolic and furanic derivatives from lignin hydrolysis and

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carbohydrate dehydration [11–13]. HTL process water with these compounds, particularly at higher HTL temperatures (> 300 °C), exhibits stronger persistence and microbial toxicity in AD systems [14,15].

Current strategies to mitigate inhibition, such as adsorption and ozone pretreatment, face limitations in scalability, cost, and potential secondary pollution [10,19–21]. One potential alternative strategy to mitigate the inhibitory effects and improve the biodegradability of HTL process water is microaeration within the anaerobic digester. Introducing small amounts of oxygen or air during AD has proven to increase microbial diversity, leading to more efficient substrate degradation and methane production [22]. Oxygen addition promotes the growth of facultative anaerobes, which can function in both aerobic and anaerobic conditions, facilitating the breakdown of complex organic compounds into simpler molecules that are more accessible to strictly anaerobic microbes [23]. Additionally, microaeration enables the limited aerobic respiration to generate extra ATP, which boosts microbial metabolism and accelerates substrate degradation [24]. However, the use of microaeration specifically for HTL process water has not been systematically explored, particularly in relation to different feedstock types.

This study aims to explore the impact of feedstock composition and microaeration on the toxicity and biodegradability of HTL process water during AD. By examining HTL process water from two dissimilar feedstocks – a food-waste proxy (*i.e.*, dog food, rich in proteins) and wheat straw (rich in lignocellulose) – this research evaluates how different chemical compositions of HTL process water affect key microbial processes in AD, particularly methanogenesis. Additionally, the study explores the potential of microaeration-acclimated biomass as a strategy to tolerate toxicity and enhance the biodegradability of HTL process water. This research provides valuable insights into optimizing AD processes for treating HTL process water and improving its environmental sustainability.

2. Materials and methods

2.1. Characteristics of HTL process water and inoculum

We generated two HTL-process-water solutions: food-waste process water (FWPW) and wheat-straw process water (WSPW) with the Aarhus University pilot-scale continuous HTL reactor at AU Viborg (Denmark) (Table S1). Dog food was selected as a proxy feedstock for protein-rich food wastes (Orlando Multico, Lidl, Denmark), while wheat straw was a model feedstock for lignocellulose-rich biomass. The HTL process conditions for both feedstocks were 350 °C for 15 min with 12% dry matter. After separating the HTL process water from the biocrude oil *via* solvent extraction, ten 10-l batches of each type of process water were shipped to our laboratory at the University of Tübingen, Germany. There, each process-water solution was filtered using 5-μm filter bags, stirred, combined in a large container, and redistributed again into 10-L bottles. The HTL-process-water solutions were stored at −20 °C until further use.

For anaerobic toxicity (AT) assays, we collected the anaerobic inoculum from an anaerobic digester that was treating animal manure at Hohenheim University (Reutlingen, Germany). For the acetoclastic methanogenic activity (AMA) assays and the biochemical methane potential (BMP) assays, we utilized biomass from an active 5.5-L laboratory AD reactor (fully anaerobic) and a microaeration-acclimated AD reactor to study the toxicity and biodegradability of HTL process water. The only operating difference between the two laboratory reactors was intermittent air dosing at 15.6 mL/min into the microaeration reactor. The biomass had acclimated to microaeration with air for one year. All assays were performed in a near-neutral pH range (6.9–7.3). The sufficient alkalinity of the inoculum made pH adjustments unnecessary.

2.2. Anaerobic toxicity (AT) assays

The classical AT assays to measure toxicity involved introducing varying chemical oxygen demand (COD) loads of HTL process water to the serum bottles containing a consistent amount of standard substrate and anaerobic inoculum [25,26]. All assays were performed in triplicate at 37 °C in 250-mL glass serum bottles with a liquid volume of 100 mL. We also performed control assays without HTL-process-water interference. The COD load of HTL process water was expressed as a percentage of the standard substrate COD. For instance, the FWPW_10% scenario (Table S2, S3) indicated that the COD added by the FWPW accounted for 10% of the standard substrate COD. WSPW_10% (Table S2, S3) indicated that the COD added by the WSPW accounted for 10% of the standard substrate COD. For the acidogenic AT assay (Table S2), we used a glucose solution (1 g/L) as the standard substrate to examine the impact of HTL process water on the acidogenesis in the AD food web [27]. We assumed that a reduction in glucose degradation rates relative to the control scenario indicated acidogenesis inhibition. We measured the glucose, short-chain carboxylic acid (SCCA), including lactate, and medium-chain carboxylic acid (MCCA) concentrations during incubation at 0, 6.5, 10.5, 22, 31, 49.5, 81.5, 105, and 127 h. For the methanogenic AT assay (Table S3), we followed the method described by Owen et al. [25], which used a mixture of acetate and propionate as the standard substrate. A reduction in the methane production rate or a prolonged lag phase, compared to the control condition (standard substrate only), was interpreted as an indication of methanogenic inhibition caused by the HTL process water. Biogas production and composition were measured daily to monitor the process.

2.3. Acetoclastic methanogenic activity (AMA) assays

Because acetoclastic methanogens are recognized as the dominant methanogenic group (*i.e.*, 80%–95% relative abundance) compared to hydrogenotrophic methanogens during AD [28,29], we performed (AMA) assays in 60-mL serum bottles to measure the toxicity of different reactants. Reactants included HTL process water from two dissimilar feedstocks (as complex reactants) and specific inhibitory compounds (as defined reactants) typically found in HTL process water. Each 60-mL serum bottle for the AMA assay contained 30 mL of liquid, consisting of anaerobic water, 15.8 mL sodium-acetate solution (2 g/L) as standard substrate, 2.8 mL biomass, and reactants, along with 0.3 mL of trace-element solution, mineral-stock solution, and Na_2S solution (10 g/L). The trace-element solution consisted of (in g/L): $C_6H_9NO_6$, 2; $MnSO_4$, 1; $FeSO_4$, 0.8; $CoCl_2$, 0.2; $ZnSO_4$, 0.2; $CuCl_2$, 0.02; $NiCl_2$, 0.02; Na_2MoO_4*$2H_2O$, 0.02; Na_2SeO_4, 0.02; and Na_2WO_4, 0.02. The mineral stock contained the following (in g/L): $NaCl_2$, 40; NH_4Cl, 50; KCl, 5; KH_2PO4, 6; $MgCl_2$, 7; and $CaCl_2$, 2.

After inoculation, the bottles were sparged with nitrogen gas for 10 min. Subsequently, the bottles were placed in a 37 °C shaking incubator (150 rpm) for 20 h. To mitigate the initial lag phase, 8 mL of sodium acetate (2 g/L) solution was supplemented to each bottle at 21 h [30]. Bottles were resealed, sparged with nitrogen gas, and returned to the incubator. The AMA was calculated and expressed as g COD-CH_4/g VSS/day according to Equations (Eqs. 1–3):

$$\text{Activity} = \frac{Slope^{*}V_{headspace}}{388^{*}VSS^{*}0.001^{*}V_{liquid}} \quad (1)$$

$$V_{headspace} = W_1 - W_2 \quad (2)$$

$$V_{liquid} = W_2 - W_0 \quad (3)$$

where *1/388* is the COD value of 1 mL CH_4 at 25 °C; *Slope* is the slope of the curve for methane composition in the headspace *versus* time (% CH_4/day); *VSS* is the volatile suspended solids at the end of the tests (g VSS/L); $V_{headspace}$ and V_{liquid} are the headspace gas volume and the liquid volume, respectively (mL); W_0, W_1, and W_2 are the weights of the blank bottle, the bottle full of distilled water, and the bottle with samples in the end, respectively (g). The density of the liquid was assumed to be 1

g/cm^3.

2.4. Biochemical methane potential (BMP) assays

We conducted BMP assays to determine the biodegradability of HTL process water from different feedstocks. Every BMP assay was performed in triplicate in 60-mL septum-capped bottles with a total liquid volume of 30 mL, consisting of HTL process water at a target COD of 1 g/L and anaerobic biomass with an inoculum/substrate (I/S) ratio of 3.5 (on a COD basis). Additionally, we added the same amount of trace-element solution, mineral-stock solution, and Na_2S solution (10 g/L), as used in the AMA assays in Section 2.3. After inoculation, the bottles were flushed with nitrogen gas and incubated at 37 °C in a shaking incubator (150 rpm). To test the efficacy of microaeration as a biodegradation strategy at two dosing intensities, we replaced nitrogen in the headspace with air to achieve oxygen partial pressures of 1% and 3% (*v/v*) (Table S4).

Biogas production was determined using the syringe method, which involved measuring the biogas volume with a glass syringe inserted directly through the septum of the bottle lid. The overpressure inside the incubated bottles pushed the piston until it balanced with atmospheric pressure. The biogas was then released into a fume hood. The measured biogas volume was normalized to standard conditions (*i.e.*, 25 °C, 101.3 kPa). The biogas volumes measured in the blank bottles were subtracted from the biogas volume measured in the experimental bottles. The cumulative methane yield curves were simulated by the modified Gompertz equation [31], as presented in Equation (Eq. 4).

$$P_t = P_m \times exp\left\{ -exp\left[\frac{R_m \times e}{P_m}(\lambda - t) + 1\right]\right\} \quad (4)$$

where P_t is the methane production volume at time t (mL); P_m is the maximum biomethane yield potential (mL); R_m is the peak methane production rate (mL/day); λ is the duration of the lag phase (day); t is the incubation period (day); e is the mathematical constant approximately equivalent to 2.7. Origin 8.5 software was used to perform a nonlinear regression analysis to determine the parameters of P_m, R_m, λ, and the correlation coefficient of determination (R^2).

2.5. Analytical methods, data analysis, and statistical analysis

The COD was analyzed according to Standard Methods [32]. SCCAs (excluding lactate) and MCCAs were determined using an Agilent 7890B gas chromatograph (GC) (Agilent Technologies Inc., Santa Clara, CA, USA) equipped with a capillary column (DB Fatwax ultra inert 30 m × 0.25 mm × 0.25 μm, length × diameter × film thickness; Agilent Technologies Inc., Santa Clara, CA, USA) and an a flame ionization detector with hydrogen gas as mobile phase, as described by Gemeinhardt et al. [33]. Glucose and lactate were determined using a high-performance liquid chromatograph (HPLC) (Shimadzu LC 20AD), which was coupled with a refractive index and UV detector (Shimadzu, Kyoto, Japan). Biogas composition was measured using a GC (SRI gas GCs, SRI Instruments, USA) equipped with a HaySep-D packed Teflon column (length 3 m, outer diameter 1/8″, SRI Instruments, Torrance, CA, USA) and a flame ionization detector for methane and a thermal conductivity detector for hydrogen and carbon dioxide with nitrogen gas as carrier gas. The GC oven was operated at isothermal conditions (*i.e.*, 70 °C). Pure methane was used as the standard for GC AMA assays, and gas samples were analyzed hourly to plot the percentage of methane in the headspace during the incubation period. Data from duplicate/triplicate measurements were expressed as mean ± standard error. All statistical analyses were performed using the OriginPro student version (OriginLab, Northampton, USA). Values of $p < 0.05$ were considered to be statistically significant.

3. Results and discussion

3.1. Methanogenesis, rather than acidogenesis, is more severely inhibited during AD of HTL process water, as shown with anaerobic toxicity (AT) assays

To determine which trophic step in the AD food web is most susceptible to HTL-process-water toxicity, we monitored glucose conversion during acidogenesis and methane production during methanogenesis for AT assays. We did not measure the activity of acetogenesis because we could not isolate this microbial activity due to overlapping activity between acidogenesis and methanogenesis [34]. The introduction of HTL process water did not markedly inhibit acidogenesis. Glucose was fully degraded within 22 h in all assays, regardless of the HTL-process-water COD load (Fig. 1a, b). Lactate was the primary intermediate produced from glucose utilization. Other SCCAs and small amounts of MCCAs were also produced, and their concentration increased during the incubation period for each scenario (FWPW_0% – WSPW_100% in Fig. 1c, d). Lactate concentrations decreased after ~31 h for all scenarios (Fig. 1a, b), while acetate and propionate concentrations increased (Fig. 1c, d), suggesting that lactate oxidation had occurred.

Lactate oxidation into acetate, propionate, and hydrogen precedes methanogenesis [35,36]. The hydrogen produced from lactate and propionate oxidation must be removed by hydrogenotrophic methanogens and maintained at low partial pressures for the oxidation reaction to occur. Lactate was completely oxidized in all scenarios involving food-waste process water (Fig. 1a). However, lactate oxidation was incomplete at higher COD loads of wheat-straw process water (*i.e.*, > WSPW_30%) (Fig. 1b), potentially due to the inhibition of lactate-utilizing bacteria, acetogens, or hydrogenotrophic methanogens. The decreased acetate accumulation rate for the scenarios > WSPW_30% (decreased slope of pink bars in Fig. 1d) compared to scenarios < WSPW_50% also indicated the inhibition of lactate conversion by higher concentrations of wheat-straw process water (Fig. 1d).

Digestion with HTL process water from different feedstocks resulted in similar inhibition trends during methanogenesis (Fig. 2). According to the time course of cumulative methane production, longer lag phases occurred at higher COD loads of HTL process water, especially in the FWPW_50%, FWPW_70%, FWPW_100%, WSPW_70%, and WSPW_100% scenarios (Fig. 2a, b). Methane production gradually increased after 6 days of incubation, possibly due to biomass acclimatization or the degradation of inhibitory compounds [17]. The total cumulative methane production was greater for higher COD loads because more COD was added and degraded.

Specific methane yields decreased at higher COD loads of the HTL process water as anticipated (Fig. 2c, d). The assays with FWPW_10% and WSPW_10% achieved the highest specific methane yields with HTL process water exposure, achieving 21.1 ± 0.57 and 20.8 ± 0.11 mL STP CH_4/g COD, respectively. However, when compared to the control assays (only acetate and propionate substrate), no differences in methane yield were detected ($p > 0.05$), indicating no apparent inhibition of methanogenesis at HTL-process-water COD loads at 10%. When COD loads exceeded 30%, methane yields were lower than the controls (Fig. 2c, d), suggesting that the process water was inhibitory or less biodegradable than the control substrate. The daily methane production rates showed earlier peak values upon adding FWPW_10% and WSPW_10%, while the peak values were delayed but increased at higher COD ratios of HTL process water (Fig. 2e, f).

To better compare the toxicity of HTL process water from different feedstocks, the time courses of specific methane yield curves (in Fig. 2c, d) were modeled using the modified Gompertz equation (Eq. 4). The simulated results were consistent with the measured impacts of higher COD loads of HTL process water on maximum methane yields (P_m), maximum daily methane production rate (R_m), and the lag phase (λ) (Table 1). According to the model, food-waste process water had a more

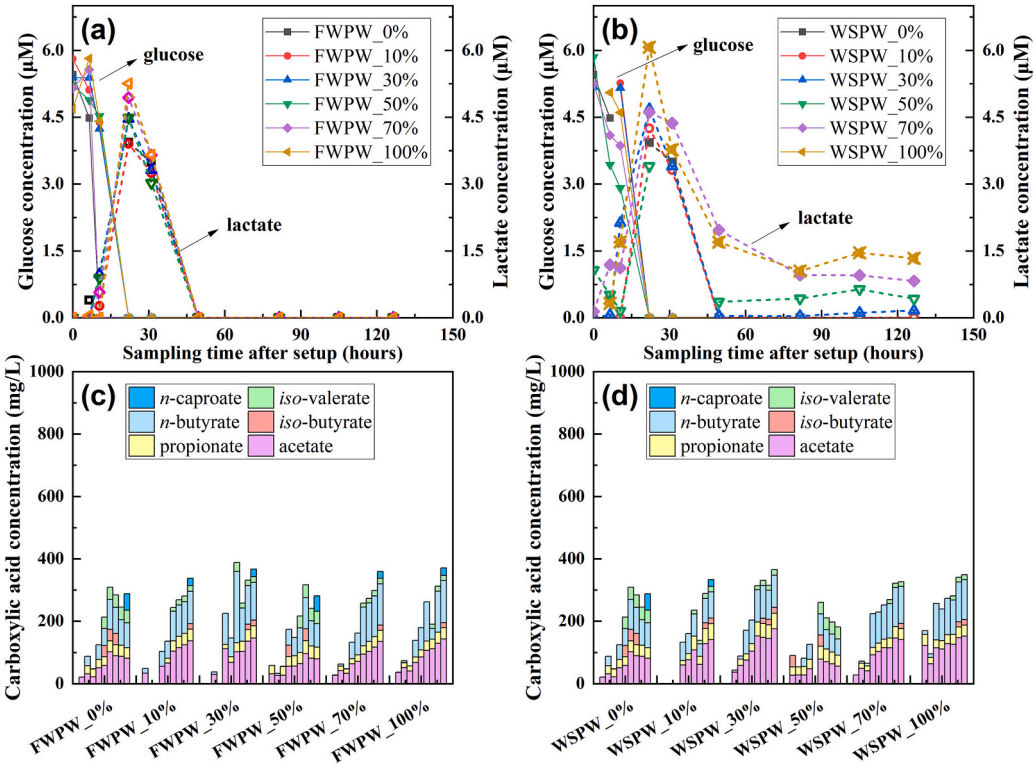


Fig. 1. Anaerobic toxicity (AT) assays for effects of food-waste process water (FWPW) and wheat-straw process water (WSPW) on acidogenesis. Glucose and lactate concentrations (**a, b**) and carboxylic acid concentrations (**c, d**). The increasing levels of HTL process water added as a feed are expressed as percentages of the standard substrate COD (*i.e.*, glucose). For each condition, sampling times were 0, 6.5, 10.5, 22, 31, 49.5, 81.5, 105, 127 h after setup. We analyzed the carboxylic acid concentration of scenarios of FWPW_0%, FWPW_50%, WSPW_0%, and WSPW_0% in the same running sequence and analyzed scenarios of FWPW_10%, FWPW_30%, FWPW_70%, FWPW_100%, WSPW_10%, WSPW_30%, WSPW_70%, and WSPW_100% in the other running sequence.

adverse effect on the maximum methane yields and maximum daily methane production rates when compared with wheat-straw process water (Table 1). Additionally, the lag phases for food-waste process water and wheat-straw process water were up to two days longer than the control (Table 1).

The complete degradation of glucose and effective production of SCCAs, including lactate, during acidogenesis (Fig. 1), alongside the prolonged lag phases observed during methanogenesis (Fig. 2 and Table 1), indicate that HTL process water exerts a more pronounced inhibitory effect on methanogenesis than on acidogenesis. Similar observations have been reported by Macêdo et al. [37], who found methanogenesis cessation and SCCA accumulation when HTL process-water concentration from sewage sludge exceeded 63% (*v*/v) in the feed. This finding is consistent with the fact that methanogens are generally more sensitive to inhibitory compounds such as ammonia nitrogen, heavy metals, antibiotics, and long-chain fatty acids [38].

3.2. HTL process water type and concentration differentially hinder acetoclastic methanogenic activity, as shown with anaerobic methanogenic activity (AMA) assays

From the analysis of AT assay results, both HTL process water types inhibited methane production when the COD load was equal to or higher than 30% of the standard substrate (Fig. 2). To understand the inhibition mechanisms better, we also conducted short-term (2-day) AMA assays to assess the toxicity to acetoclastic methanogens from different concentrations of food-waste process water and wheat-straw process water. In addition, we also assessed the toxicity to acetoclastic methanogens from specific inhibitory compounds, which are common in process water. The second acetate dosing in each assay, which was administered after approximately one day of incubation, primarily contributed to methane production due to its fast kinetics (and thus not the COD from the added process water or compounds) [42]. The initial incubation period of one day allowed for the adaptation and stabilization of biomass in response to acetate as a substrate. The pH level for all bottles was within the neutral range of 6.9–7.3 throughout the two-day period, thanks to the high alkalinity of the inoculum.

Food-waste process water and wheat-straw process water influenced acetoclastic methanogenic activity differently at elevated concentrations ($p < 0.01$ in Fig. 3a). Food-waste process water did not significantly reduce acetoclastic methanogenic activity when the COD load was up to 2 g/L ($p > 0.05$ in Fig. 3a). Notably, the toxicity of wheat-straw process water was more severe with the increased concentrations ($p < 0.01$ in Fig. 3a). The half maximal inhibitory concentrations (IC_{50}), which corresponds to 50% inhibition, was 2.5 g COD/L for wheat-straw process water, but considerably higher for food-waste process water and outside of our tested concentrations (> 5 g COD/L in Table 2). The molecular structure of inhibitors can be an important factor in inhibiting acetoclastic methanogenic activity to varying degrees, and therefore we measured this activity (Fig. 3b, c, d). Here, we found, at the same concentration ranges, that aromatic compounds (no N or O in the ring structure) with a hydroxyl group (4-hydroxy-acetophenone and phenol) showed a higher toxicity to acetoclastic methanogenic activity than 2,4,6-trimethoxyacetophenone without a hydroxyl group ($p < 0.0001$ in Fig. 3b). For N-heteroaromatic compounds, the number of heteroatoms correlated with the degree of toxicity to acetoclastic methanogens. For instance, methyl-pyrazine (2 N in the ring) reduced the acetoclastic methanogenic activity more severely than methyl-pyridine ($p < 0.05$) and 3-hydroxypyridine (1 N in the ring) ($p < 0.0001$ in Fig. 3c).

Additionally, for the non-aromatic heterocyclic compounds, the toxicity of furfural (with O as the heteroatom) was significantly influenced by the concentration ($p < 0.05$), whereas the toxicity of

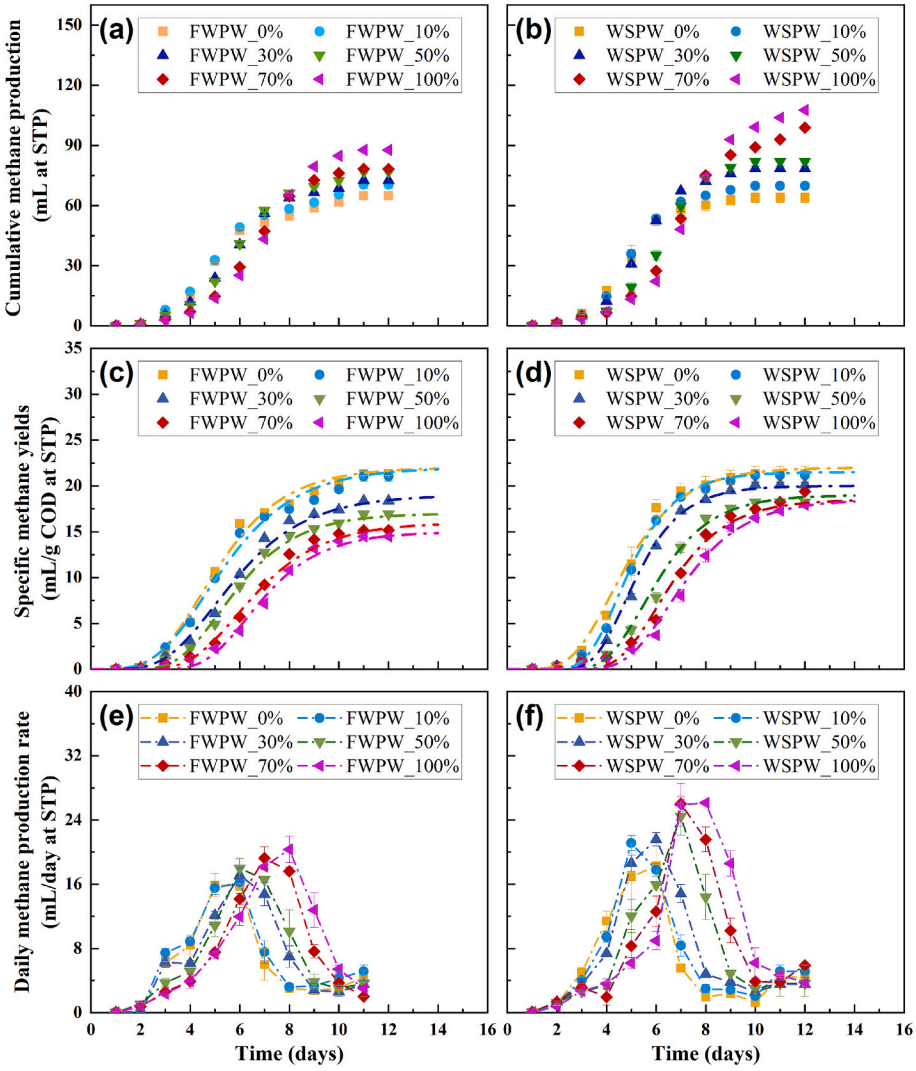


Fig. 2. Anaerobic toxicity (AT) assays for effects of food-waste process water (FWPW) and wheat-straw process water (WSPW) on methanogenesis: Cumulative methane yields (mL at STP) (**a, b**), specific methane yields (mL/g COD at STP) (**c, d**), and daily methane production rate (mL/day) (**e, f**). The increasing levels of HTL process water added as a feed are expressed as percentages of the standard substrate COD (*i.e.*, the mixture of sodium acetate (10 g/L) and propionic acid (6.69 mL/L)). Data for each experimental condition are based on replicate experiments using separate bottles. Statistical analysis was performed using Two-Way ANOVA, and the results are presented as mean ± standard deviation (data points + error bars). Some error bars may not be fully visible due to overlapping data points; detailed numerical ranges can be found in Table S5.

pyrrolidone (with N as the heteroatom) was not significantly affected by the concentration ($p > 0.05$ in Fig. 3d). In general, higher toxicity to acetoclastic methanogens was found for the individual compounds that can be anticipated to be more abundant in wheat-straw process water, such as the aromatic compounds and furfural without N in their aromatic rings, compared to food-waste process water, such as the N-heteroaromatic compounds and 2-pyrrolidone with N (Fig. 3b, c, d, and Table 2). This finding is consistent with the higher toxicity that we observed for wheat-straw process water than food-waste process water.

A greater decline in acetoclastic methanogenic activity was observed with wheat-straw process water compared to food-waste process water upon increasing its concentration (Fig. 3). This difference in toxicity can be attributed to the varying composition of organic compounds in the two types of process water, particularly the relative abundance of phenols, furans, and nitrogen-containing heterocycles, which are commonly found in HTL process water [39,40]. Here, it was noticed that compound functional groups (*e.g.*, hydroxyl *vs.* methoxy groups), aromaticity, and the number and type of heteroatoms in the ring structures play a key role in determining toxicity (Table 2). Notably, compounds typically derived from lignocellulose-rich feedstocks (*e.g.*, phenols, furans) [12,43] showed stronger inhibition of acetoclastic methanogens, and thereby explaining the higher overall toxicity observed in wheat-straw process water.

3.3. HTL process water from protein-rich feedstock is less biodegradable than that from lignocellulose-rich feedstock, as shown with biochemical methane potential (BMP) assays

For the BMP assays, HTL process water was added to the bottles at a COD concentration of 1 g/L to measure biodegradability at a very mild toxicity level. Indeed, AMA assays showed that adding 1 g COD/L of food-waste process water and wheat-straw process water resulted in only 8.5% and 20% reduction in acetoclastic methanogenic activity, respectively (Fig. 3a). Other indications of relatively low toxicity levels for HTL process water at a concentration of 1 g COD/L are: **(1)** the methane yield increased rapidly in the initial phase without a lag phase (Fig. 4a); and **(2)** all SCCAs, excluding lactate, remained below the detection level (~ 0.47 mM) during the BMP-assay period of 30 days.

The biodegradability of food-waste process water and wheat-straw process water were considerably different, as indicated by their

Table 1
Experimental and kinetic model parameters simulated by the modified Gompertz equation for food-waste process water (FWPW) and wheat-straw process water (WSPW).

Substrates	Measured maximum methane yields ($mLCH_4/gCOD_{added}$)	Kinetic model parameters		
		P_m ($mLCH_4/gCOD_{added}$)	R_m ($mLCH_4/gCOD_{added}$/day)	λ (day)
FWPW_0%	21.4 ± 0.20	22	12	2.6
FWPW_10%	21.1 ± 0.57	22	11	2.6
FWPW_30%	18.4 ± 0.08	19	9.5	3
FWPW_50%	16.9 ± 0.20	17	10	3.5
FWPW_70%	15.2 ± 0.19	16	8.5	4
FWPW_100%	14.5 ± 0.04	15	9	4.6
WSPW_0%	22.6 ± 0.60	22	14	2.6
WSPW_10%	20.8 ± 0.11	21.5	17	3.2
WSPW_30%	19.8 ± 0.04	20	16	3.5
WSPW_50%	18.1 ± 0.07	19	12.8	4
WSPW_70%	18.1 ± 0.02	18.5	12	4.5
WSPW_100%	17.1 ± 0.01	18.5	11	4.8

P_m: simulated maximum methane yields; R_m: simulated maximum daily methane production rate; λ: lag phase.

respective specific methane yields (Fig. 4a). After 30 days of anaerobic digestion, the specific methane yields were 130 ± 0.13 and 189 ± 1.5 mL CH_4/g COD for food-waste process water and wheat-straw process water, respectively ($p < 0.0001$ in Fig. 4a). A lower biodegradability of food-waste process water than wheat-straw process water was also shown by a lower methane content in the biogas during the initial phase when the methane content was rising and not at steady state yet ($p < 0.0001$ in Fig. 4b). The lower biodegradability of food-waste process water could be due to the presence of recalcitrant N-heteroaromatic compounds, such as pyridines and pyrazines, which are produced from protein during HTL [19,45].

The apparent contradiction between the toxicity and biodegradability of wheat-straw process water and food-waste process water underscores the critical role of feedstock composition in determining HTL process water treatability (Fig. 3a and Fig. 4a). While wheat-straw process water showed higher toxicity, its superior methane yield compared to food-waste process water (189 *vs.* 130 mL CH_4/g COD) likely originates from the preferential degradation of oxygenated compounds (*e.g.*, ketones, furans) by acetogens and facultative bacteria [10]. In contrast, the recalcitrance of food-waste process water can be traced to N-heterocyclic compounds (*e.g.*, pyrazines), which resist anaerobic degradation due to their stable aromatic rings and unreactive functional groups [19]. Supporting this observation, Chen et al. [12,46] reported that HTL process water from rice straw (comparable to our lignocellulose-rich wheat straw) had a higher methane potential of 217

Table 2
The 50% inhibition concentration (IC_{50}) of process water and typical organic inhibitors on acetoclastic methanogenic activity as shown with the AMA assay.

Inhibitors	50% inhibition concentration (IC_{50})
Food-waste process water	Outside our range, likely >5 g COD/L
Wheat-straw process water	2.5 g COD/L
4-Hydroxy-acetophenone	0.42 g/L
Furfural	0.94 g/L
Phenol	1.1 g/L
2,4,6-trimethoxyacetophenone	2.0 g/L
2-Pyrrolidinone	1.8 g/L
Methyl-pyrazine	Outside our range, likely >4 g/L
Methyl-pyridine	Outside our range, likely >6 g/L
3-Hydroxypyridine	Outside our range, likely >7 g/L

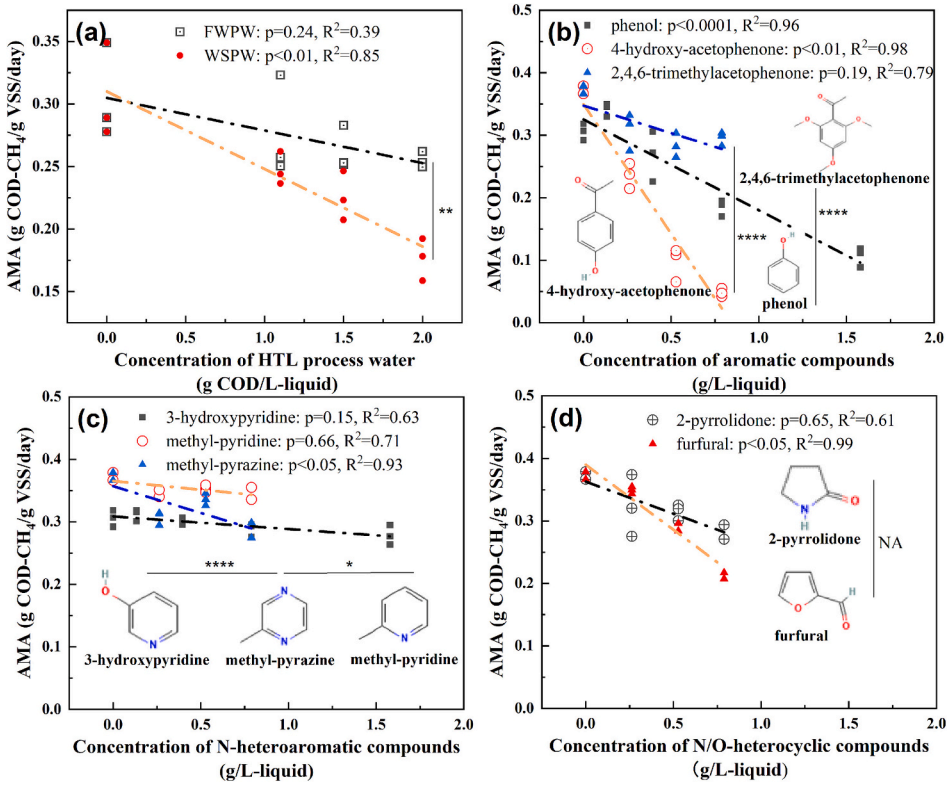


Fig. 3. Acetoclastic methanogenic activity (AMA) assays for the effect of different reactants on acetoclastic methanogenic activity (expressed as g COD-CH_4/g VSS/day). Reactants include the increasing concentrations of food-waste process water (FWPW) and wheat-straw process water (WSPW) (**a**) and specific inhibitory compounds, including aromatic compounds (**b**), N-heteroaromatic compounds (**c**), and N/O-heterocyclic compounds (**d**). We conducted One-Way ANOVA for each reactant (*p*-value in the legend) to assess the effect of elevating concentrations on AMA. We also assessed the significance between AMA and different reactants across concentrations by using Two-Way ANOVA (p-value with asterisks in the figure, * indicates $p < 0.05$, **** indicates <0.0001, and NA is no significant difference).

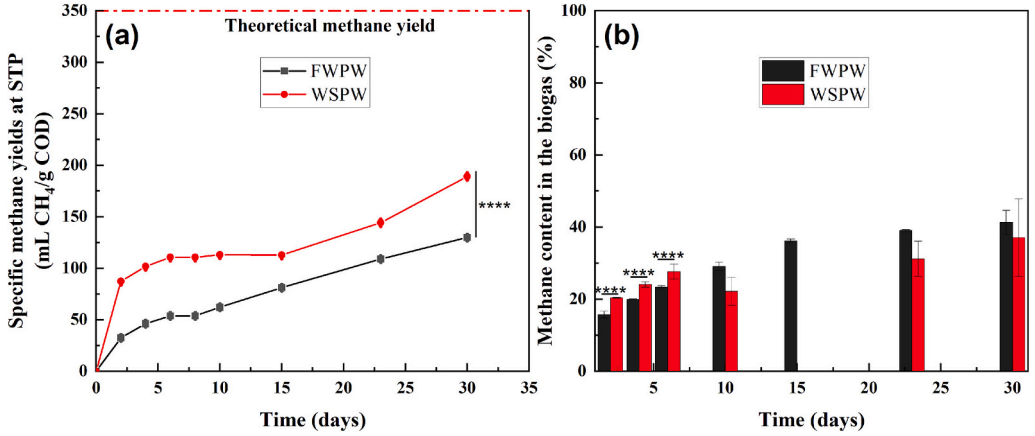


Fig. 4. Biochemical methane potential (BMP) assays for specific methane yields of HTL process water (**a**) and methane content in the biogas (**b**) from food-waste process water (FWPW) and wheat-straw process water (WSPW). The methane production from the anaerobic inoculum was subtracted to calculate the specific methane yields from the two process water types. Data for each experimental condition are based on duplicate experiments using separate bottles. The results are presented as mean ± standard error (data points + error bars) and compared with the Paired-Sample *t*-Test (p-value with asterisks in the figure, **** indicates <0.0001). Some error bars may not be fully visible due to overlapping data points; detailed numerical ranges can be found in Table S6.

mL CH_4/g COD than that from dewatered sewage sludge (similar to our protein-rich food-waste proxy), which yielded 136 mL CH_4/g COD under the same HTL conditions (320 °C, 30 min).

3.4. Microaeration-acclimated biomass enhances the biodegradability of one HTL process water type, as shown with the biochemical methane potential (BMP) assays

We tested the tolerance of microaeration-acclimated biomass to the two HTL process water types with AMA assays without applying any oxygen. Food-waste process water at concentrations lower than 2 g COD/L did not show a notable toxicity difference between microaeration-acclimated biomass and anaerobic biomass ($p > 0.05$ in Fig. 5a). When exposed to wheat-straw process water, microaeration-acclimated biomass exhibited a lower acetoclastic methanogenic activity compared with the anaerobic biomass ($p > 0.0001$ in Fig. 5b), showing a slight reduction of acetoclastic methanogen tolerance to wheat-straw process water with *prior* microaeration acclimation.

In addition to toxicity tolerance, we tested the ability of microaeration-acclimated biomass to biodegrade the two HTL-process-water types while varying the oxygen partial pressure in the headspace (Fig. 6). It is evident that microaeration-acclimated biomass consistently produced more methane from food-waste process water compared to anaerobic biomass ($p < 0.0001$ in Fig. 6a, c, e), even without the addition of oxygen (Fig. 6a). For wheat-straw process water, however, the specific methane yields were not significantly different ($p > 0.05$ in Fig. 6b, d, f). We also observed that for both HTL process water types, increasing the O_2 partial pressures lowered the specific methane yields for each biomass, even for the microaeration-acclimated biomass (Fig. 6). These findings emphasize the need for feedstock-specific strategies for AD of HTL process water. In particular, for protein-rich HTL feedstocks, such as food waste, conventional toxicity mitigation methods (*e.g.*, adsorption or dilution) may be insufficient, as the primary barrier is poor biodegradability rather than acute toxicity. Instead, microbial adaptation strategies, such as microaeration acclimation, offer a more promising route. Microaeration-acclimated biomass improved the methane yield from food-waste process water by 38% (no oxygen addition; Fig. 6a), likely due to the breakdown of persistent N-heterocyclic compounds. The introduction of limited oxygen has been reported to enrich facultative and aerobic microbes (*e.g.*, *Pseudomonas* spp.) [47], which are capable of partially oxidizing or transforming recalcitrant N-heterocyclic compounds into more biodegradable intermediates [48]. In contrast, wheat-straw process water, although initially more toxic, was more readily degradable regardless of microaeration. Therefore, for

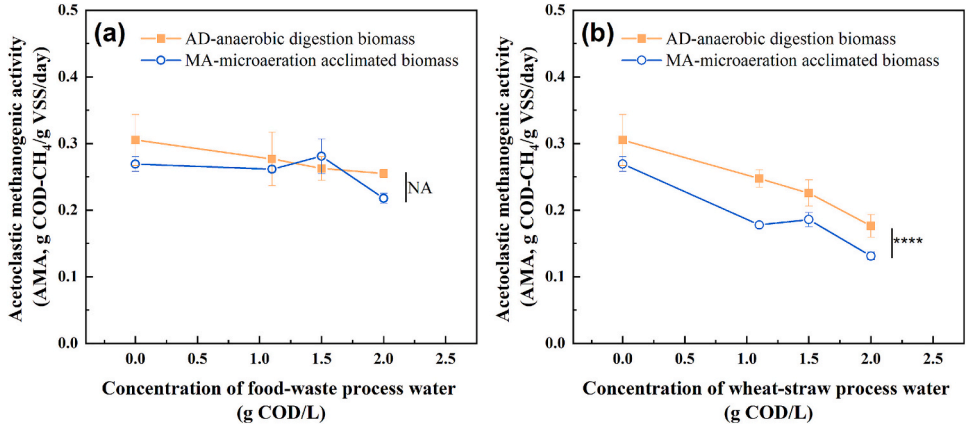


Fig. 5. Acetoclastic methanogenic activity (AMA) assays for the effect on the activity of anaerobic digestion (AD) and microaeration-acclimated (MA) biomass with different concentrations of food-waste process water (FWPW) (**a**) and wheat-straw process water (WSPW) (**b**). All AMA bottles were kept strictly anaerobic. Data for each experimental condition are based on triplicate experiments using separate bottles. Statistical analysis was performed using Two-Way ANOVA, and the results are presented as mean ± standard deviation (data points + error bars) (p-value with asterisks in the figure, **** indicates <0.0001, NA indicates no significant difference).

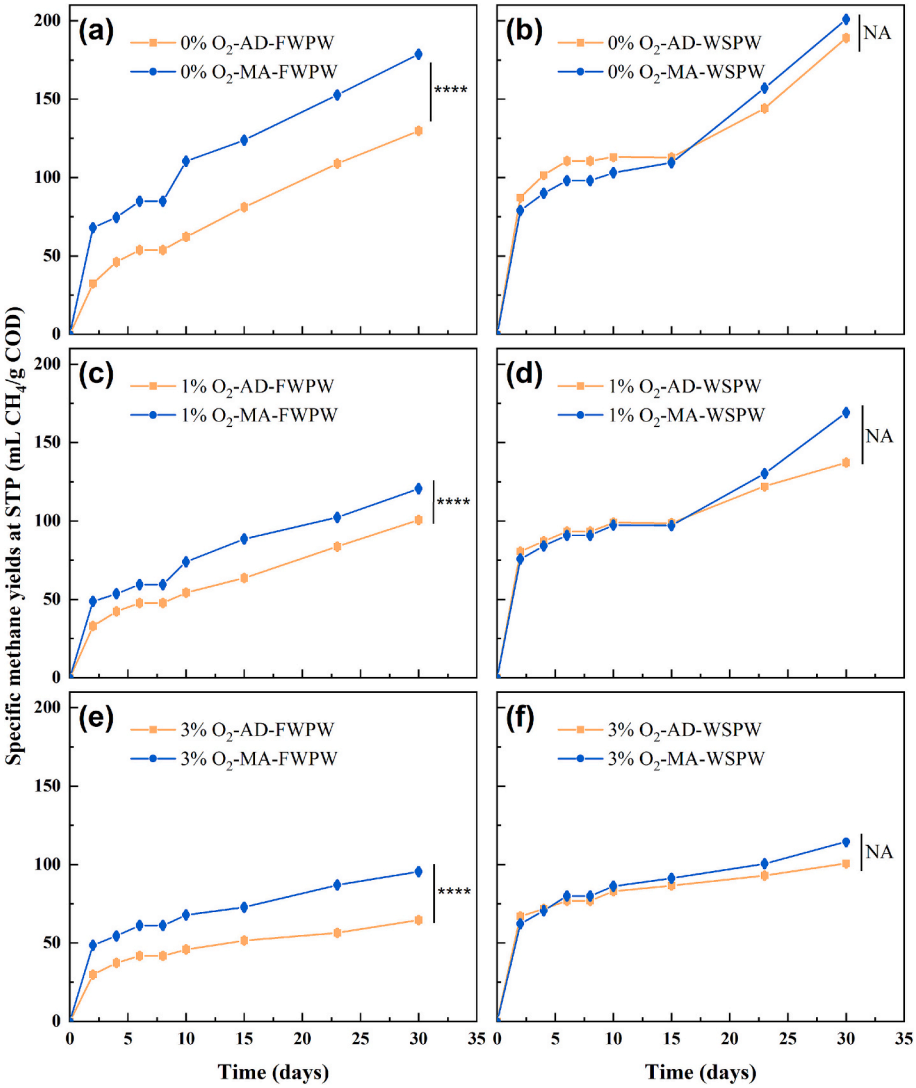


Fig. 6. Biochemical methane potential (BMP) assays for the effects of anaerobic digestion (AD) and microaeration-acclimated (MA) biomass on biodegradation of food-waste process water (FWPW) (**a, c, e**) and wheat-straw process water (WSPW) (**b, d, f**) at the same concentration of 1 g COD/L-liquid and varying oxygen partial pressures in the headspace. Data for each experimental condition are based on duplicate experiments using separate bottles. Statistical analysis was performed using Two-Way ANOVA, and the results are presented as mean ± standard error (data points + error bars) (p-value with asterisks in the figure, **** indicates <0.0001, NA indicates no significant difference). Some error bars may not be fully visible due to overlapping data points; detailed numerical ranges can be found in Table S7.

lignocellulose-rich HTL feedstocks, strategies should instead focus on targeted removal of key inhibitors (*e.g.*, phenol adsorption) to alleviate methanogenesis inhibition and maximize energy recovery.

4. Conclusions and future perspectives

This study highlights the critical influence of feedstock composition on the AD performance of HTL process water. While process water derived from protein-rich food waste exhibited lower acute toxicity, it showed limited methane production due to the persistence of recalcitrant nitrogenous compounds. In contrast, process water from lignocellulose-rich wheat straw, though more toxic, was more biodegradable, suggesting different rate-limiting factors depending on the feedstock. Notably, microaeration acclimation improved methane yields from food-waste process water, suggesting a clear advantage of using microbial adaptation compared to the conventional AD system for HTL process water from protein-rich feedstocks.

Future research should prioritize the molecular-level identification of inhibitory compounds across diverse HTL feedstocks and operating conditions. Additionally, long-term studies on microaeration-acclimated systems under continuous operation are necessary to validate their stability and scalability. These efforts will be essential to optimize HTL-AD integration and unlock the full potential of wet waste valorization in sustainable bioenergy systems.

CRediT authorship contribution statement

Mei Zhou: Writing – original draft, Visualization, Investigation, Conceptualization. **Joseph G. Usack:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Aidan Mark Smith:** Writing – review & editing, Resources. **Largus T. Angenent:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The first author used AI-assisted technologies to revise the initial manuscript draft to provide clearer and more concise English expressions. After using this tool, the first author reviewed and edited the

content again and took full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Largus T. Angenent reports financial support was provided by Alexander von Humboldt Foundation. Mei Zhou reports financial support was provided by China Scholarship Council. Joseph G. Usack reports financial support was provided by German Israeli Foundation for Scientific Research and Development. Aidan Mark Smith reports financial support was provided by Poul Due Jensen Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.165540.

Data availability

Data will be made available on request.

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